Their size varies from 1 micron to 50 meter. Algae may be unicellular and microscopic (Eg. *Chlamydomonas*) or multicellular and macroscopic (Eg. *Sargassum*). Unicellular algae exhibits variety of shapes (i.e., spherical, rod, spindle), where as multicellular algae are in the form of filaments and branches. In this section we will study about unicellular algae (Chlamydomonas) alone. Multicellular algae are explained in detail in the next chapter.

16.4.1 Cell structure of Chlamydomonas

Chlamydomonas is a simple, unicellular, motile fresh water algae. They are oval, spherical or pyriform in shape. The pyriform (pear shape) is a common one found in ponds, ditches and water tanks. They have a narrow anterior end and a broad posterior end.

The cell is surrounded by a thin and firm cell wall made of cellulose. The cytoplasm is seen in between the cell membrane and the chloroplast. The cell contains large dark nucleus lying inside the cavity of the cup shaped chloroplast. The anterior part of the cell bears two flagella which helps in locomotion. Two contractile vacuoles are seen at the base of each flagellum. The anterior side of the chloroplast contains a tiny red coloured eyespot. *Chlamydomonas* exhibits sexual and asexual modes of reproduction.

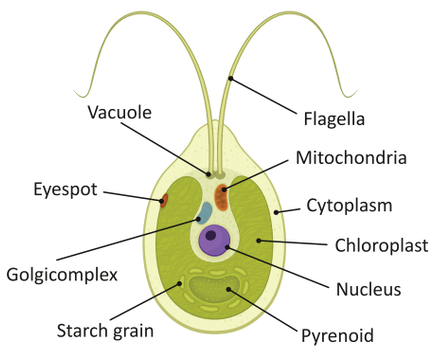


Figure 16.7 Structure of Chlamydomonas

16.5 Protozoa

A protozoan (In Greek, 'protos' means first and 'zoan' means animal) is a single-celled eukaryote. They are included under the kingdom **Protista**. The study of protozoa is called **Protozoology**. They are found in ponds, ocean, in moist soil, and in the cells and tissues of plants and animals causing diseases. They range from 2 to 200 microns. Protozoans have specialized organelles. These organelles are used for movement, feeding, and other functions. The types of protozoans are as follows:

Ciliates: Presence of cilia for locomotion.

Eg. *Paramecium*

Flagellates: Presence of flagella for locomotion.
Eg. *Euglena*

Pseudopods: Presence of pseudopodia for locomotion. Eg. *Amoeba*

Sporozoans: Parasites.
Eg. *Plasmodium*

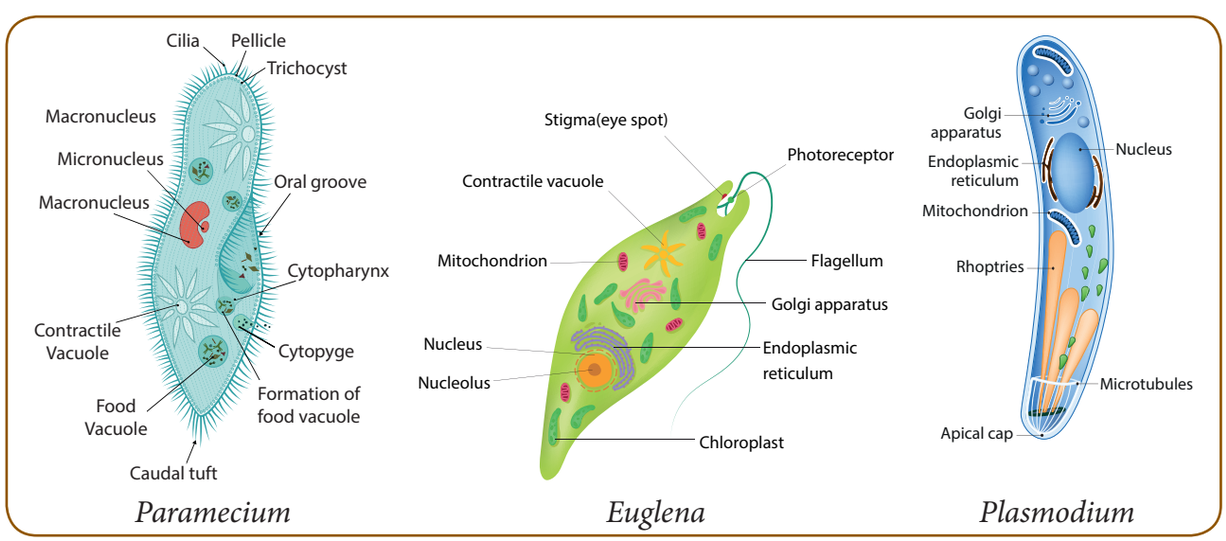


Figure 16.8 Common Protozoans

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